

A Biased Pentagonal Cut and the Bound $\delta_c \leq 0.78876$

Abstract

For a graph G , let $t(G)$ and $b(G)$ be the maximum numbers of edges in a triangle-free and a bipartite subgraph of G , respectively, and let δ_c be the asymptotic minimum-degree threshold forcing $t(G) = b(G)$. Balogh, Keevash and Sudakov proved $3/4 \leq \delta_c < 0.791$ and conjectured $\delta_c = 3/4$. We prove

$$\delta_c \leq \frac{19719}{25000} = 0.78876.$$

The proof follows their deletion-and-structure method. The new ingredient is a biased random cut for a cyclic five-part partition. The remaining calculation is reduced to five exact sign checks for univariate rational functions, certified by Sturm's theorem.

1 Introduction

Erdős asked when the largest triangle-free and largest bipartite subgraphs of a graph must have the same number of edges [2]. For an n -vertex graph G , write $t(G)$ and $b(G)$ for these two maximum edge counts. Balogh, Keevash and Sudakov [1] defined the asymptotic threshold δ_c , proved $3/4 \leq \delta_c < 0.791$, and conjectured that the lower bound is sharp.

We improve their upper bound.

Theorem 1. *For all sufficiently large n , every n -vertex graph G with*

$$\delta(G) \geq \frac{19719}{25000}n$$

satisfies $t(G) = b(G)$. Consequently,

$$\delta_c \leq \frac{19719}{25000} = 0.78876.$$

The proof retains the framework of [1]: take a maximum triangle-free subgraph, delete vertices below a suitable relative-degree threshold, and apply Jin's homomorphism theorem together with the weighted estimates of Balogh–Keevash–Sudakov. The resulting core has minimal Andrásfai type F_1, F_2, F_3 or F_4 . The F_1, F_3, F_4 branches are excluded by explicit inequalities. In the $F_2 = C_5$ branch, the symmetric pentagonal cut of [1] is replaced by the cyclic Bernoulli vector

$$\left(0, 1, 0, \frac{18347}{25000}, \frac{18347}{25000}\right).$$

The only computer-assisted step is Lemma 2. Its exact verification code is reproduced in Appendix A and uses rational arithmetic throughout.

2 Structural tools and the biased cut

Let F_j denote the j -regular Andrásfai graph on $3j - 1$ vertices; in particular, $F_1 = K_2$ and $F_2 = C_5$. A graph has F_j -type if it admits a homomorphism to F_j .

Proposition 1 (Jin; Balogh–Keevash–Sudakov). *Let Γ be a triangle-free graph on m vertices with $\delta(\Gamma) \geq \gamma m$.*

1. *If $\gamma > 5/14$, then Γ has F_4 -type.*
2. *Suppose $j \in \{2, 3, 4\}$ is minimal such that Γ has F_j -type, and write $e(\Gamma) = \beta m^2$. Then*

$$\begin{aligned} j = 2: \quad & \gamma \leq \frac{2}{5}, \quad \beta \leq 5\gamma^2 - 4\gamma + 1, \quad \frac{1}{m^2} \sum_{i \in \mathbb{Z}_5} |V_i|^2 \leq 30\gamma^2 - 24\gamma + 5, \\ j = 3: \quad & \gamma \leq \frac{3}{8}, \quad \beta \leq 36\gamma^2 - 27\gamma + \frac{21}{4}, \\ j = 4: \quad & \gamma \leq \frac{4}{11}, \quad \beta \leq \frac{297}{4}\gamma^2 - 54\gamma + 10. \end{aligned}$$

Here V_i are the five parts of the C_5 -homomorphism in the case $j = 2$.

3. *If Γ is a bipartite subgraph of a graph G and M edges of G are incident with $V(G) \setminus V(\Gamma)$, then G has a bipartite subgraph with at least $e(\Gamma) + M/2$ edges.*

Proof. The first assertion is [1, Theorem 2.1], with parameter $j = 4$, and is due to Jin [3]. The estimates in the second assertion are [1, Lemma 2.2]. The third assertion is [1, Lemma 3.2]. $\square$

Lemma 1 (Biased pentagonal cut). *Suppose a graph has a partition $V_0 \cup \dots \cup V_4$, with indices modulo 5. Let e_0, e_1, e_2 count the edges whose endpoints lie, respectively, in the same part, in adjacent parts, and in parts at cyclic distance two. Then the graph has a bipartite subgraph with at least*

$$\pi_0 e_0 + \pi_1 e_1 + \pi_2 e_2$$

edges, where

$$\pi_0 = \frac{122062591}{781250000}, \quad \pi_1 = \frac{1205737591}{1562500000}, \quad \pi_2 = \frac{2}{5}.$$

In particular, $\pi_0 < \pi_2 < \pi_1$.

Proof. Put $\tau = 18347/25000$ and

$$(r_0, r_1, r_2, r_3, r_4) = (0, 1, 0, \tau, \tau).$$

Choose $s \in \mathbb{Z}_5$ uniformly. Independently for every vertex $v \in V_i$, place v on the first side of a cut with probability r_{i-s} . If two vertices lie at cyclic distance $k \in \{0, 1, 2\}$, their average separation probability is

$$\frac{1}{5} \sum_{j \in \mathbb{Z}_5} (r_j(1 - r_{j+k}) + (1 - r_j)r_{j+k}).$$

For $k = 0, 1, 2$, respectively, these probabilities are

$$\frac{4\tau(1 - \tau)}{5}, \quad \frac{2(1 + 2\tau - \tau^2)}{5}, \quad \frac{2}{5}.$$

Substituting $\tau = 18347/25000$ gives the displayed values of π_0, π_1, π_2 . Linearity of expectation yields a cut of the asserted size. $\square$

3 Proof of Theorem 1

Proof. Set

$$d = \frac{19719}{25000}, \quad g_0 = \frac{4d-1}{6} = \frac{13469}{37500}, \quad \varepsilon_n = n^{-1/4}.$$

Suppose the theorem is false. Then, along an infinite sequence of orders $n \rightarrow \infty$, there are graphs G_n with $\delta(G_n) \geq dn$ and $b(G_n) < t(G_n)$. Let $H_n \subseteq G_n$ be a maximum-edge triangle-free subgraph and write

$$e(H_n) = \rho_n n^2.$$

Every graph has a cut containing at least half of its edges, while $e(G_n) \geq dn^2/2$. Hence

$$\frac{d}{4} < \rho_n \leq \frac{1}{4},$$

where the upper bound is Mantel's theorem.

Define

$$g_n = \frac{4\rho_n - (2d-1)^2}{6-8d+8\rho_n}, \quad \rho(g) = \frac{(6-8d)g + (2d-1)^2}{4-8g}. \quad (1)$$

The denominator in the definition of g_n is positive, and $\rho_n = \rho(g_n)$. Direct calculation gives

$$\rho(g_0) = \frac{d}{4}, \quad \rho\left(\frac{d}{2}\right) = \frac{1}{4}, \quad \rho'(g) = \frac{27888961}{312500000(2g-1)^2} > 0.$$

Therefore

$$g_0 < g_n \leq \frac{d}{2} < \frac{2}{5}. \quad (2)$$

Set

$$\gamma_n = g_n - \varepsilon_n.$$

Since $g_0 - 5/14 = 533/262500 > 0$, we have $\gamma_n > 5/14$ for all sufficiently large n .

Starting from H_n , whenever the current graph has k vertices and contains a vertex of degree less than $\gamma_n k$, delete such a vertex. Let Γ_n be the final graph, and write

$$m_n = |V(\Gamma_n)| = \alpha_n n, \quad e(\Gamma_n) = \beta_n m_n^2.$$

The sum of the degrees deleted is at most

$$\gamma_n \sum_{k=m_n+1}^n k = \frac{\gamma_n}{2} (n(n+1) - m_n(m_n+1)).$$

Consequently,

$$\beta_n \alpha_n^2 \geq \rho_n - \frac{\gamma_n}{2} (1 - \alpha_n^2) - \frac{\gamma_n}{2n} (1 - \alpha_n). \quad (3)$$

Moreover,

$$2\rho(g) - g = -\frac{(25000g - 7219)^2}{312500000(2g-1)} > 0 \quad \left(g_0 \leq g \leq \frac{d}{2}\right).$$

This continuous function is bounded away from zero on the displayed compact interval. Hence the right-hand side of (3) is positive when $\alpha_n = 0$ and n is large, so Γ_n is nonempty. By construction,

$$\delta(\Gamma_n) \geq \gamma_n m_n.$$

Proposition 1 implies that Γ_n has F_4 -type. Let $j_n \in \{1, 2, 3, 4\}$ be its minimal Andrásfai type.

After passing to a subsequence, we may assume that $j_n = j$ is constant and that

$$g_n \rightarrow g, \quad \alpha_n \rightarrow \alpha, \quad \beta_n \rightarrow \beta.$$

Here $g_0 \leq g \leq d/2$, $\gamma_n \rightarrow g$, and $\rho_n \rightarrow \rho(g)$. Taking limits in (3) gives

$$\left(\beta - \frac{g}{2}\right) \alpha^2 \geq \rho(g) - \frac{g}{2} > 0. \quad (4)$$

In particular, $\alpha > 0$ and $\beta - g/2 > 0$.

The bipartite-core branch $j = 1$. Suppose first that Γ_n is bipartite. If $\alpha_n = 1$, then no vertex was deleted, so $\Gamma_n = H_n$ is bipartite, contradicting $b(G_n) < e(H_n)$. Hence $\alpha_n < 1$.

Let $S_n = V(G_n) \setminus V(\Gamma_n)$ and let M_n be the number of edges of G_n incident with S_n . Since an edge inside S_n is counted twice in the degree sum over S_n ,

$$M_n = \sum_{v \in S_n} d_{G_n}(v) - e_{G_n}(S_n) \geq (n - m_n)dn - \binom{n - m_n}{2}. \quad (5)$$

By Proposition 1(3),

$$b(G_n) \geq e(\Gamma_n) + \frac{M_n}{2}.$$

Combining this with $b(G_n) < e(H_n)$ and the deletion estimate before normalization yields

$$\frac{\gamma_n}{2} (n(n+1) - m_n(m_n+1)) > \frac{M_n}{2}.$$

Using (5), dividing by $(n - m_n)n > 0$, and rearranging gives

$$(1 - 2\gamma_n)\alpha_n < 1 + 2\gamma_n - 2d - \frac{1 - 2\gamma_n}{n}. \quad (6)$$

On the other hand, Γ_n is triangle-free, so $\beta_n \leq 1/4$. Equation (3) therefore gives

$$(1 - 2\gamma_n)\alpha_n^2 \geq 4\rho_n - 2\gamma_n - \frac{2\gamma_n}{n}(1 - \alpha_n). \quad (7)$$

Since $1 - 2\gamma_n > 0$, (6) and (7) imply

$$\Delta_n(\gamma_n) < \frac{2\gamma_n(1 - 2\gamma_n)}{n}, \quad (8)$$

where

$$\Delta_n(x) = (1 - 2x)(4\rho_n - 2x) - (1 + 2x - 2d)^2.$$

Indeed, (6) gives

$$\alpha_n < \frac{1 + 2\gamma_n - 2d}{1 - 2\gamma_n},$$

while (7) gives

$$(1 - 2\gamma_n)\alpha_n^2 \geq 4\rho_n - 2\gamma_n - \frac{2\gamma_n}{n};$$

combining these two inequalities gives (8).

The definition of g_n is equivalent to $\Delta_n(g_n) = 0$, and direct differentiation gives

$$\Delta'_n(x) = -(6 - 8d + 8\rho_n).$$

Since $\gamma_n = g_n - \varepsilon_n$,

$$\Delta_n(\gamma_n) = (6 - 8d + 8\rho_n)\varepsilon_n > 6(1 - d)n^{-1/4}. \quad (9)$$

But $0 < \gamma_n < 1/2$, so the right-hand side of (8) is at most $1/(4n)$. This contradicts (9) for large n . Hence $j \neq 1$.

The branches $j = 3$ and $j = 4$. Suppose $j = 3$. Proposition 1(2) gives

$$g \leq \frac{3}{8}, \quad \beta \leq U_3(g) := 36g^2 - 27g + \frac{21}{4}.$$

The function $\rho(g)$ is increasing, while U_3 is non-increasing on $(-\infty, 3/8]$. Moreover,

$$\rho(g_0) - U_3(g_0) = \frac{210187}{312500000} > 0.$$

Thus $\beta < \rho(g)$, which contradicts (4), because $\alpha \leq 1$ and $\beta - g/2 > 0$.

If $j = 4$, then

$$g \leq \frac{4}{11}, \quad \beta \leq U_4(g) := \frac{297}{4}g^2 - 54g + 10.$$

The function U_4 is non-increasing on $(-\infty, 4/11]$, and

$$\rho(g_0) - U_4(g_0) = \frac{8683037}{625000000} > 0.$$

Again $\beta < \rho(g)$, contradicting (4). Therefore $j = 2$.

The C_5 -core branch $j = 2$. Set

$$B_5(g) = 5g^2 - 4g + 1, \quad c(g) = 30g^2 - 24g + 5.$$

Proposition 1(2) yields

$$\beta \leq B_5(g), \quad \limsup_{n \rightarrow \infty} \frac{1}{m_n^2} \sum_{i \in \mathbb{Z}_5} |V_{i,n}|^2 \leq c(g), \quad (10)$$

where $V_{0,n}, \dots, V_{4,n}$ are the five parts of a C_5 -homomorphism of Γ_n .

From (4), $\beta - g/2 > 0$. Since $\alpha \leq 1$,

$$\beta - \frac{g}{2} \geq \left(\beta - \frac{g}{2}\right) \alpha^2 \geq \rho(g) - \frac{g}{2}.$$

Hence $B_5(g) \geq \beta \geq \rho(g)$. Lemma 2(1) therefore gives

$$g_0 \leq g < g_1 := \frac{36633}{100000}. \quad (11)$$

Also,

$$B_5(g) - \frac{g}{2} = 5 \left(g - \frac{2}{5}\right) \left(g - \frac{1}{2}\right) > 0 \quad (g < g_1 < 2/5).$$

Combining (4) with $\beta \leq B_5(g)$ gives

$$\alpha^2 \geq u(g) := \frac{\rho(g) - g/2}{B_5(g) - g/2}. \quad (12)$$

By Lemma 2(2), $u(g) \geq 3/5$, and therefore

$$\alpha > \frac{1}{2}. \quad (13)$$

For each n , put $T_n = V(\Gamma_n)$ and $S_n = V(G_n) \setminus T_n$, and define

$$p_n = \frac{2e_{G_n}(S_n)}{n^2}, \quad q_n = \frac{2e_{G_n}(T_n)}{n^2}, \quad r_n = \frac{e_{G_n}(T_n, S_n)}{n^2}.$$

After passing to a further subsequence, assume $p_n \rightarrow p$, $q_n \rightarrow q$, and $r_n \rightarrow r$. Summing degrees over T_n and S_n , and using the elementary edge-capacity bounds, gives

$$\begin{aligned} q + r &\geq d\alpha, & p + r &\geq d(1 - \alpha), \\ 0 \leq p &\leq (1 - \alpha)^2, & 0 \leq q &\leq \alpha^2, & 0 \leq r &\leq \alpha(1 - \alpha). \end{aligned} \quad (14)$$

Let $e_{\ell,n}$ be the number of edges of $G_n[T_n]$ of cyclic type $\ell \in \{0, 1, 2\}$ with respect to the partition $V_{0,n}, \dots, V_{4,n}$. Then

$$e_{0,n} + e_{1,n} + e_{2,n} = \frac{q_n n^2}{2}.$$

Every edge of Γ_n has type one, so (3) gives

$$\frac{e_{1,n}}{n^2} \geq \frac{e(\Gamma_n)}{n^2} \geq \rho_n - \frac{\gamma_n}{2}(1 - \alpha_n^2) - \frac{\gamma_n}{2n}(1 - \alpha_n). \quad (15)$$

Furthermore, by (10),

$$\frac{e_{0,n}}{n^2} \leq \frac{1}{2n^2} \sum_{i \in \mathbb{Z}_5} |V_{i,n}|^2 \leq \frac{1}{2} c(\gamma_n) \alpha_n^2. \quad (16)$$

Apply Lemma 1 inside T_n and place every vertex of S_n independently on either side with probability $1/2$. There is a cut of G_n with at least

$$\frac{p_n n^2}{4} + \frac{r_n n^2}{2} + \pi_0 e_{0,n} + \pi_1 e_{1,n} + \pi_2 e_{2,n}$$

edges. Since $\pi_2 = 2/5$, this is

$$n^2 \left(\frac{p_n}{4} + \frac{r_n}{2} + \frac{q_n}{5} \right) + (\pi_1 - \frac{2}{5}) e_{1,n} - (\frac{2}{5} - \pi_0) e_{0,n}.$$

Using (15) and (16), then taking lower limits, we obtain

$$\liminf_{n \rightarrow \infty} \frac{b(G_n)}{n^2} \geq \frac{p}{4} + \frac{q}{5} + \frac{r}{2} + \left(\pi_1 - \frac{2}{5} \right) h(g, \alpha) - \frac{\frac{2}{5} - \pi_0}{2} c(g) \alpha^2, \quad (17)$$

where

$$h(g, \alpha) = \rho(g) - \frac{g}{2}(1 - \alpha^2). \quad (18)$$

It remains to minimize the part of (17) depending on p, q, r . From (14),

$$r \geq r_1 := d(1 - \alpha) - (1 - \alpha)^2, \quad r \geq r_2 := d\alpha - \alpha^2.$$

Since $\alpha > 1/2$,

$$r_1 - r_2 = (1 - d)(2\alpha - 1) > 0,$$

so $r \geq r_1$. Also, the two degree constraints imply

$$\begin{aligned} \frac{p}{4} + \frac{q}{5} + \frac{r}{2} &\geq \frac{d(1 - \alpha) - r}{4} + \frac{d\alpha - r}{5} + \frac{r}{2} \\ &= \frac{d(1 - \alpha)}{4} + \frac{d\alpha}{5} + \frac{r}{20} \end{aligned}$$

$$\geq \frac{d(1-\alpha)}{4} + \frac{d\alpha}{5} + \frac{r_1}{20}.$$

Equality is attained at

$$p_* = (1-\alpha)^2, \quad r_* = d(1-\alpha) - (1-\alpha)^2, \quad q_* = \alpha^2 - (1-d)(2\alpha-1). \quad (19)$$

Indeed, $p_* + r_* = d(1-\alpha)$ and $q_* + r_* = d\alpha$, while

$$\begin{aligned} r_* &= (1-\alpha)(d+\alpha-1) \geq 0, \\ \alpha(1-\alpha) - r_* &= (1-d)(1-\alpha) \geq 0, \\ \alpha^2 - q_* &= (1-d)(2\alpha-1) \geq 0. \end{aligned}$$

and $q_* = d\alpha - r_* \geq \alpha(d+\alpha-1) > 0$. Thus (19) is feasible and is the minimizer.

Define

$$\begin{aligned} \Phi(g, \alpha) &= \frac{p_*}{4} + \frac{q_*}{5} + \frac{r_*}{2} + \left(\pi_1 - \frac{2}{5}\right) \left(\rho(g) - \frac{g}{2}(1-\alpha^2)\right) \\ &\quad - \frac{\frac{2}{5} - \pi_0}{2} c(g)\alpha^2 - \rho(g). \end{aligned} \quad (20)$$

By Lemma 2, $\Phi(g, \cdot)$ is concave and

$$\Phi(g, 1) > 0, \quad \Phi(g, \sqrt{u(g)}) > 0 \quad (g_0 \leq g < g_1).$$

Since (12) gives $\sqrt{u(g)} \leq \alpha \leq 1$, concavity yields

$$\Phi(g, \alpha) > 0.$$

The right-hand side of (17) is therefore at least $\rho(g) + \Phi(g, \alpha) > \rho(g)$. On the other hand,

$$\frac{b(G_n)}{n^2} < \rho_n \longrightarrow \rho(g),$$

a contradiction. This completes the proof. $\square$

4 Exact algebraic verification

Retain the constants and functions introduced above, and let Φ be given by (20). Since p_*, q_*, r_* are quadratic or linear in α , one may write

$$\Phi(g, \alpha) = A(g)\alpha^2 + B\alpha + C(g).$$

Exact expansion gives

$$B = \frac{5281}{250000} > 0. \quad (21)$$

Lemma 2. *Let*

$$J = \left[\frac{13469}{37500}, \frac{d}{2} \right], \quad I = \left[\frac{13469}{37500}, \frac{36633}{100000} \right].$$

The following statements hold.

1. *The function $F(g) = B_5(g) - \rho(g)$ is strictly decreasing on J , and*

$$F\left(\frac{36633}{100000}\right) = -\frac{106078137}{26734000000000} < 0.$$

Consequently, $F(g) \geq 0$ and $g \in J$ imply $g < 36633/100000$.

2. On I ,

$$u(g) \geq \frac{3}{5}, \quad A(g) < 0, \quad \Phi(g, 1) > 0.$$

3. Put

$$E(g) = A(g)u(g) + C(g).$$

On I ,

$$E(g) < 0, \quad B^2u(g) - E(g)^2 > 0.$$

Consequently,

$$\Phi(g, \sqrt{u(g)}) > 0 \quad (g \in I).$$

Proof. For the first assertion, direct differentiation gives

$$F'(g) = \frac{M(g)}{312500000(2g-1)^2},$$

where

$$M(g) = 12500000000g^3 - 17500000000g^2 + 8125000000g - 1277888961.$$

Moreover,

$$M'(g) = 625000000(2g-1)(30g-13).$$

On J we have $2g-1 < 0$ and $30g-13 < 0$, so $M'(g) > 0$. Since

$$M(d/2) = -\frac{286726408041}{10000} < 0,$$

it follows that $M(g) < 0$ throughout J , hence $F'(g) < 0$. The displayed endpoint value of F is an exact substitution.

For the remaining assertions, substitute the definitions of $d, \tau, \rho, B_5, c, p_*, q_*, r_*$ into (20). After cancellation, the denominators of the five rational functions below are positive on I ; their only nonconstant factors are powers of $1-2g$ and $2-5g$. Exact Sturm counts for their numerators give

quantity	numerator degree	roots in I	sign at g_0
$u - 3/5$	3	0	+
$-A$	2	0	+
$\Phi(g, 1)$	3	0	+
$-E$	4	0	+
$B^2u - E^2$	8	0	+

Thus all five displayed quantities are positive throughout I . The exact rational reconstruction, denominator checks, endpoint signs and Sturm counts are carried out by the code in Appendix A.

Finally, $B > 0$ by (21), $u(g) > 0$, $E(g) < 0$, and

$$B^2u(g) > E(g)^2.$$

Therefore $B\sqrt{u(g)} > |E(g)| = -E(g)$, and hence

$$\Phi(g, \sqrt{u(g)}) = A(g)u(g) + B\sqrt{u(g)} + C(g) = E(g) + B\sqrt{u(g)} > 0.$$

□

Remark 1. The use of limits in the F_2, F_3, F_4 branches is only a compactness device. Every contradiction there has a strictly positive limiting margin. The F_1 branch is treated at finite n : the exact tangency surplus is of order $n^{-1/4}$, whereas the deletion error is $O(n^{-1})$.

A Exact verification code

The following program reconstructs all rational functions from the displayed definitions. It requires `sympy`; all arithmetic and all Sturm counts are exact.

```
1  #!/usr/bin/env python3
2  """Exact verification for the bound delta_c <= 19719/25000.
3
4  All computations use exact rational arithmetic. Polynomial root counts are
5  performed by Sturm's theorem through SymPy's Poly.count_roots method.
6  """
7  import sympy as sp
8
9  g, alpha, x = sp.symbols('g alpha x', real=True)
10 d = sp.Rational(19719, 25000)
11 tau = sp.Rational(18347, 25000)
12 g0 = sp.Rational(13469, 37500)
13 g1 = sp.Rational(36633, 100000)
14 gmax = d / 2
15
16 assert g0 == (4*d - 1)/6
17 assert g0 < g1 < gmax < sp.Rational(2, 5) < sp.Rational(1, 2)
18
19 rho = sp.factor(((6 - 8*d)*g + (2*d - 1)**2)/(4 - 8*g))
20 assert rho.subs(g, g0) == d/4
21 assert rho.subs(g, gmax) == sp.Rational(1, 4)
22 assert sp.factor(sp.diff(rho, g)) == sp.Rational(27888961, 312500000)/(2*g - 1)**2
23 assert sp.simplify(2*rho - g + (25000*g - 7219)**2/(312500000*(2*g - 1))) == 0
24
25 # Tangency used in the bipartite-core branch. Here rho is held fixed while
26 # x is varied.
27 Delta = sp.factor((1 - 2*x)*(4*rho - 2*x) - (1 + 2*x - 2*d)**2)
28 assert sp.factor(Delta.subs(x, g)) == 0
29 assert sp.simplify(sp.diff(Delta, x) + (6 - 8*d + 8*rho)) == 0
30
31 # F3 and F4 endpoint margins.
32 U3 = 36*g**2 - 27*g + sp.Rational(21, 4)
33 U4 = sp.Rational(297, 4)*g**2 - 54*g + 10
34 m3 = sp.factor((rho - U3).subs(g, g0))
35 m4 = sp.factor((rho - U4).subs(g, g0))
36 assert m3 == sp.Rational(210187, 312500000)
37 assert m4 == sp.Rational(8683037, 625000000)
38
39 # The biased pentagonal Bernoulli cut.
40 rvec = [sp.Rational(0), sp.Rational(1), sp.Rational(0), tau, tau]
41 def sep(a, b):
42     return sp.expand(a + b - 2*a*b)
43
44 p0 = sp.factor(sum(sep(a, a) for a in rvec)/5)
45 p1 = sp.factor(sum(sep(rvec[i], rvec[(i + 1) % 5]) for i in range(5))/5)
46 p2 = sp.factor(sum(sep(rvec[i], rvec[(i + 2) % 5]) for i in range(5))/5)
47 assert p0 == sp.Rational(122062591, 781250000)
48 assert p1 == sp.Rational(1205737591, 1562500000)
49 assert p2 == sp.Rational(2, 5)
50 assert p0 < p2 < p1
51
52 # C5 structural quantities and the completion linear program.
53 B5 = 5*g**2 - 4*g + 1
54 c5 = 30*g**2 - 24*g + 5
55 u = sp.factor((rho - g/2)/(B5 - g/2))
56 assert sp.simplify(B5 - g/2 - 5*(g - sp.Rational(2, 5))*(g - sp.Rational(1, 2))) == 0
57
58 pstar = (1 - alpha)**2
59 rstar = d*(1 - alpha) - (1 - alpha)**2
60 qstar = alpha**2 - (1 - d)*(2*alpha - 1)
61 assert sp.factor(pstar + rstar - d*(1 - alpha)) == 0
62 assert sp.factor(qstar + rstar - d*alpha) == 0
63 assert sp.simplify(rstar - (d*alpha - alpha**2) - (1 - d)*(2*alpha - 1)) == 0
64 assert sp.simplify(alpha*(1 - alpha) - rstar - (1 - d)*(1 - alpha)) == 0
65 assert sp.simplify(alpha**2 - qstar - (1 - d)*(2*alpha - 1)) == 0
```

```

66 assert sp.factor(sp.Rational(1, 2) - sp.Rational(1, 4) - sp.Rational(1, 5)) == sp.Rational(1, 20)
67
68 h = rho - g*(1 - alpha**2)/2
69 cut = sp.factor(
70     pstar/4 + rstar/2 + p2*qstar/2
71     + (p1 - p2)*h - (p2 - p0)*c5*alpha**2/2
72 )
73 Phi = sp.factor(cut - rho)
74 Palpha = sp.Poly(sp.expand(Phi), alpha)
75 A = sp.factor(Palpa.coeff_monomial(alpha**2))
76 B = sp.factor(Palpa.coeff_monomial(alpha))
77 C = sp.factor(Palpa.coeff_monomial(1))
78 assert B == sp.Rational(5281, 250000)
79
80 # Feasible C5 parameters lie below g1.
81 F = sp.factor(B5 - rho)
82 M = sp.together(sp.diff(F, g)).as_numer_denom()[0]
83 assert sp.simplify(sp.diff(M, g) - 625000000*(2*g - 1)*(30*g - 13)) == 0
84 assert M.subs(g, gmax) == -sp.Rational(286726408041, 10000)
85 assert F.subs(g, g1) == -sp.Rational(106078137, 26734000000000)
86
87
88 def sturm_positive(expr, lo, hi, expected_degree):
89     """Certify expr>0 on [lo,hi] by exact numerator/denominator root counts."""
90     num, den = sp.cancel(expr).as_numer_denom()
91     if den.subs(g, lo) < 0:
92         num, den = -num, -den
93     num_poly = sp.Poly(sp.expand(num), g)
94     den_poly = sp.Poly(sp.expand(den), g)
95     assert num_poly.degree() == expected_degree
96     assert num_poly.eval(lo) > 0
97     assert den_poly.eval(lo) > 0
98     assert num_poly.count_roots(lo, hi) == 0
99     assert den_poly.count_roots(lo, hi) == 0
100
101
102 # The five exact sign checks used in the proof.
103 sturm_positive(u - sp.Rational(3, 5), g0, g1, 3)
104 sturm_positive(-A, g0, g1, 2)
105 sturm_positive(Phi.subs(alpha, 1), g0, g1, 3)
106 E = sp.factor(A*u + C)
107 Q = sp.factor(B**2*u - E**2)
108 sturm_positive(-E, g0, g1, 4)
109 sturm_positive(Q, g0, g1, 8)
110
111 print('Exact verification passed.')
112 print('d =', d, '=', float(d))
113 print('tau =', tau, '=', float(tau))
114 print('g interval =', g0, 'to', g1)
115 print('cut probabilities =', p0, p1, p2)
116 print('F3/F4 margins =', m3, m4)
117 print('linear-program slope in r =', sp.Rational(1, 20))
118 print('Sturm checks: u-3/5>0, A<0, Phi(g,1)>0, E<0, B^2u-E^2>0.')

```

Listing 1: Exact verification of the algebraic claims.

B Expected verification output

Running Listing 1 produces

```

Exact verification passed.
d = 19719/25000 = 0.78876
tau = 18347/25000 = 0.73388
g interval = 13469/37500 to 36633/100000
cut probabilities = 122062591/781250000 1205737591/1562500000 2/5
F3/F4 margins = 210187/312500000 8683037/625000000
linear-program slope in r = 1/20
Sturm checks: u-3/5>0, A<0, Phi(g,1)>0, E<0, B^2u-E^2>0.

```

References

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- [2] P. Erdős, On some problems in graph theory, combinatorial analysis and combinatorial number theory, in: *Graph Theory and Combinatorics, Cambridge 1983*, Academic Press, London, 1984, 1–17.
- [3] G. Jin, Triangle-free graphs with high minimal degrees, *Combin. Probab. Comput.* **2** (1993), 479–490.